Ameyassh Nagarajan

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Education

Oregon State University — Corvallis, OR

Ph.D. & M.Eng. in Artificial Intelligence and Crop Science, GPA: 3.87/4.00 — *Sep 2022 – Dec 2028*

- **Dissertation research:** Developing low-cost, edge-deployable computer vision systems for automated crop monitoring and pest management.
- **Relevant coursework:** Artificial Intelligence, Algorithms and Data Structures, Deep Learning, Natural Language Processing, Computer Vision, High-Performance Computing for Deep Learning

University of Mumbai — Mumbai, Maharashtra, India

Bachelor of Engineering in Information Technology, CGPA: 8.16/10.00 — *Jun 2018 – May 2022*

- **Undergraduate thesis:** Natural language processing applications in competitive gaming strategy optimization.
- **Relevant coursework:** Database Management Systems, Operating Systems, Software Engineering, Machine Learning

Research Interests

Computer Vision & Perception; Multimodal Machine Learning & Sensor Fusion; Multimodal Remote Sensing; Edge AI & Real-Time Inference; Object Detection; Internet of Things (IoT); Camera Systems & Interfacing; Precision Agriculture

Research Experience

Graduate Research Assistant — Oregon State University

Corvallis, OR — *May 2024 – Present*

Slug Identification in Grass Seed Farms, Willamette Valley, Oregon

- Designed and deployed **48 semi-autonomous camera systems** and **12 weather monitoring units** built on **Raspberry Pi 4**, **ESP32**, and Pi Camera NoIR modules, plus humidity and light sensors, a wind vane, and an anemometer, with a one-week battery life.
- Developed an **optical-flow-based pest-detection network** to detect slug movement in grass seed fields.
- Paired network predictions with observations from the weather monitoring units to predict and understand slug emergence, and ran **multivariate analysis** on how weather conditions influence emergence.

Grass Seed Species Identification and Sorting

- Developed a camera system for grass seed species identification to support seed analysts, with custom camera software for **Raspberry Pi** and **NVIDIA Jetson Orin Nano**.
- Built the full image acquisition pipeline, tuning camera parameters and lighting, with file management and **server-side inference**.
- Integrated multiple cameras with the Jetson Orin Nano and benchmarked camera performance and acquisition speed using **libcamera**, **GStreamer**, and **GenICam**.
- Recognized by the Oregon Seed Association and featured by the OSU College of Science and Capital Press.

DIRT: Data-driven Inference of Rootzone and Terrain

- Collaborating with the OPEnS Lab under Dr. John Selker at OSU on a project funded and supported by **Stevens Water**.
- Developed a sensor suite combining **RGB, NIR, LWIR, and point hyperspectral (HSI)** sensors, and calibrated the RGB, NIR, and LWIR cameras.
- Developed a **transformer-based multimodal classification model** to detect multiple stressors on a golf course.
- Developed a method to minimize infrared leakage in RGB sensors.

Grady Sensor Web Application

- Built a full-stack web application for the **Grady Sensor** grass-seed moisture monitoring system using **Node.js**, **React**, and **Stripe**, supporting technology transfer and commercial deployment.

Undergraduate Research Assistant — SIES Graduate School of Technology

Mumbai, India — *Jun 2021 – Mar 2022*

- Conducted research in **Natural Language Processing** under **Prof. Samundiswary**, investigating machine learning applications in strategic decision-making systems.
- Formulated the draft selection problem in DOTA 2 as a sequence completion task, developing a recommendation system for team composition optimization.
- Implemented and evaluated multiple neural architectures including Continuous Bag of Words (CBOW) and **Long Short-Term Memory (LSTM)** networks, achieving statistically significant improvements in recommendation accuracy.

- Published findings at the **2022 6th International Conference on Trends in Electronics and Informatics (ICOEI)**, demonstrating the applicability of NLP techniques to real-time strategic decision support systems.

Teaching Experience

Teaching Assistant – Precision Agriculture Oregon State University
Corvallis, OR *Spring 2025*

- Assisted with instruction for a graduate-level course covering machine learning, computer vision, and image processing applications in agricultural systems.
- Developed and delivered lectures on image processing techniques and machine learning methodologies for agricultural applications.
- Developed **3 labs**, exercises, and interactive **Jupyter notebooks** to help students at all levels understand the basics of deep learning.
- Evaluated student assignments and mentored students on advanced topics in computer vision and deep learning, and on the use of GIS in agriculture.

Machine Learning Instructor iD Tech Camps, University of Washington
Seattle, WA *Jun 2023 – Aug 2023*

- Delivered instruction in **Python programming** and **machine learning fundamentals**, covering model development workflows in **TensorFlow** with emphasis on data preprocessing, **Convolutional Neural Networks (CNNs)** for image classification, and **Recurrent Neural Networks (RNNs)** for sequence modeling.
- Developed pedagogical materials including programming exercises, project scaffolds, and instructional presentations tailored to diverse student proficiency levels.
- Enhanced curriculum by identifying content gaps and proposing improvements to instructional quality and technical rigor.

Publications

Y. Mohammed, S. Srinivasan, S. Iyer, and **A. Nagarajan**, "Defense of the Ancients (DOTA 2)-Draft Recommendation System," *2022 6th International Conference on Trends in Electronics and Informatics (ICOEI)*, Tirunelveli, India, 2022, pp. 13–17, doi: 10.1109/ICOEI53556.2022.9776822.

Presentations

Invited Speaker, Oregon Seed Association Grass Workshop, Salem, OR Apr 2026
- Presented advances to the seed species identification system and led a live demonstration.

Invited Speaker, Oregon Seed Association Mid-Winter Meeting, Lebanon, OR Jan 2026
- Presented research on seed species identification using computer vision.

Oral Presentation, ASABE 2025 Annual International Meeting, Toronto, Canada 2025
- **A. Nagarajan** et al., "Digitalizing Seed Purity Testing: Identifying Species of Seeds using Low-cost Computer Vision and Artificial Intelligence System."

Mentoring Experience

Henry Dalyrymple, Undergraduate Student; next: Starlink, SpaceX (2026) Jan 2025 – Jun 2026

Ajay Krishna Alawarswamy, Master's Student Jun 2026 – Present

Press and Public Coverage

Capital Press, "Delegating grass seed scrutiny to computers" Feb 2025

OSU College of Science, "Helping Oregon farmers thrive with smarter seed testing solutions" Jan 2025

Awards and Honors

Kent Wiley Memorial Scholarship, Oregon Seed Association ($2,500) 2025

Technical Skills

Programming Languages: Python, C/C++, Java, SQL, R, JavaScript, Bash, LaTeX
Frameworks and Libraries: PyTorch, TensorFlow, CUDA, OpenMP, OpenGL, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Gensim, PyQt
Edge & Embedded Vision: NVIDIA Jetson, libcamera, Picamera2, GStreamer, GenICam
Web Technologies: HTML/CSS, Flask, Node.js, React
Development Tools: Git, PostgreSQL, SQLite